CLASS 9 PHYSICS

SOUND

Complete Concepts • Formulae • Examples • Practice Questions • Solutions

Chapter Snapshot
Learn how sound is produced and propagated, characteristics of sound waves, frequency, time period, amplitude, wavelength, speed, loudness, pitch, quality, reflection of sound, echo, reverberation, and the human ear.

1. Production of Sound

Sound is produced by **vibrating objects**. When an object vibrates, it makes the particles of the surrounding medium vibrate and transfers energy from one particle to the next.

Examples: vibrating guitar strings, vocal cords, tuning fork and drum membrane.

Key Point
Sound needs a **material medium** for propagation. Therefore, sound cannot travel through vacuum.

2. Propagation of Sound

In air, sound generally travels as a **longitudinal wave**. The particles of the medium vibrate back and forth parallel to the direction of wave travel.

Compression: region of high pressure and high density.
Rarefaction: region of low pressure and low density.

Sound transfers **energy**, not matter, from one place to another.

3. Characteristics of a Sound Wave

Amplitude (A)

Maximum displacement of a vibrating particle from its mean position. Larger amplitude generally produces louder sound.

Time Period (T)

Time taken for one complete vibration.

Frequency (f)

Number of complete vibrations per second. SI unit: **hertz (Hz)**.

$$f = 1 / T$$

Wavelength (λ)

Distance between two successive compressions or two successive rarefactions.

Speed of Wave (v)

$$v = f\lambda$$

SI units: speed = m/s, wavelength = m, frequency = Hz.

4. Loudness, Pitch and Quality

Loudness depends mainly on amplitude. Greater amplitude → louder sound.
Pitch depends on frequency. Higher frequency → shriller sound.
Quality (timbre) helps distinguish sounds produced by different sources even when they have the same pitch and loudness.

Remember
Amplitude controls loudness; frequency controls pitch. Do not confuse the two.

5. Speed of Sound

The speed of sound depends on the medium and conditions such as temperature. In general, sound travels faster in solids than in liquids and faster in liquids than in gases.

Approximate speed of sound in air at room conditions is about **340 m/s**.

6. Reflection of Sound

Sound waves can reflect from hard surfaces. The laws of reflection of sound are similar to those for light: the angle of incidence equals the angle of reflection.

Applications include megaphones, horns, stethoscopes, soundboards and SONAR.

7. Echo

An **echo** is the repetition of a sound heard after reflection from a distant surface. For a distinct echo, the reflected sound should reach the ear after about **0.1 s or more**.

$$\text{Distance} = \text{Speed} \times \text{Time}$$

For echo distance, sound travels to the obstacle and back, so:

$$\text{Distance of obstacle} = (v \times t) / 2$$

8. Reverberation

The persistence of sound in a hall due to repeated reflections is called **reverberation**. Excessive reverberation makes sound unclear.

Auditoriums use curtains, carpets, cushioned seats and sound-absorbing materials to control excessive reverberation.

9. Range of Hearing

The human ear can generally hear frequencies from about **20 Hz to 20,000 Hz (20 kHz)**.

Infrasonic: below 20 Hz.
Audible: 20 Hz to 20 kHz.
Ultrasonic: above 20 kHz.

Ultrasound is used in medical imaging, cleaning delicate machine parts and detecting internal flaws.

10. SONAR

SONAR stands for Sound Navigation And Ranging. It uses ultrasonic waves and their reflection to determine the distance or depth of underwater objects.

$$\text{Distance} = (\text{Speed of sound in water} \times \text{Time taken}) / 2$$

11. Human Ear

The outer ear collects sound waves. Vibrations travel through the ear canal to the eardrum. The middle-ear bones transmit and amplify vibrations, while the inner ear converts them into nerve signals that are interpreted by the brain.

Solved Examples

Example 1: Frequency

A tuning fork makes 200 vibrations in 2 s. Find frequency.

f = vibrations/time = 200/2 = **100 Hz**.

Example 2: Time period

Find the time period of a wave with frequency 50 Hz.

T = 1/f = 1/50 = **0.02 s**.

Example 3: Wave speed

A wave has frequency 500 Hz and wavelength 0.68 m. Find speed.

$v = f\lambda = 500 \times 0.68 =$ **340 m/s**.

Example 4: Echo

An echo is heard 2 s after shouting. Take speed of sound = 340 m/s. Find distance of reflecting surface.

$d = vt/2 = 340 \times 2 / 2 =$ **340 m**.

Example 5: Wavelength

Sound speed is 330 m/s and frequency is 660 Hz. Find wavelength.

$\lambda = v/f = 330/660 =$ **0.5 m**.

Practice Questions

1. How is sound produced?
2. Why can sound not travel through vacuum?
3. What are compressions and rarefactions?
4. Define amplitude, frequency and time period.
5. State the relation between frequency and time period.
6. A source vibrates 500 times in 10 s. Find frequency.
7. A wave has frequency 250 Hz and wavelength 1.2 m. Find speed.
8. Differentiate between loudness and pitch.
9. What is an echo? Write the condition for hearing a distinct echo.
10. An echo is heard after 4 s. Find the distance of the obstacle if sound speed is 340 m/s.
11. Define reverberation and state two methods to reduce it.
12. State the audible range of human hearing.
13. What are infrasonic and ultrasonic sounds?
14. Explain the principle and working idea of SONAR.
15. Describe briefly how the human ear helps us hear sound.

Answers & Solutions

1. Sound is produced by vibrating objects.

2. Sound needs particles of a material medium to transfer vibrations; vacuum has no such particles.

3. Compressions are high-pressure, high-density regions; rarefactions are low-pressure, low-density regions.

4. Amplitude: maximum displacement. Frequency: vibrations per second. Time period: time for one vibration.

5. **$f = 1/T$** and **$T = 1/f$**.

6. $f = 500/10 =$ **50 Hz**.

7. $v = f\lambda = 250 \times 1.2 =$ **300 m/s**.

8. Loudness depends mainly on amplitude; pitch depends on frequency.

9. Echo is reflected sound heard separately. The delay should be about **0.1 s or more**.

10. $d = vt/2 = 340 \times 4 / 2 =$ **680 m**.

11. Reverberation is persistence of sound due to repeated reflections. Reduce it using curtains, carpets, cushioned seats or sound-absorbing materials.

12. Approximately **20 Hz to 20,000 Hz**.

13. Infrasonic: below 20 Hz. Ultrasonic: above 20 kHz.

14. SONAR sends ultrasonic waves and detects reflected waves. Time delay is used to calculate distance or depth.

15. The outer ear collects sound, the eardrum vibrates, middle-ear bones transmit vibrations, and the inner ear converts them into nerve signals.

Quick Revision

Sound	Produced by vibrating objects
Medium	Sound cannot travel through vacuum
Frequency	$f = 1/T$
Time Period	$T = 1/f$
Wave Speed	$v = f\lambda$
Loudness	Mainly depends on amplitude
Pitch	Depends on frequency
Echo distance	$d = vt/2$
Audible range	20 Hz to 20 kHz
Ultrasound	Frequency above 20 kHz
SONAR	Uses reflection of ultrasonic waves

Exam Tip: For numericals, write Given → Formula → Substitution → Answer with unit. For echo and SONAR, remember to divide by 2 because sound travels to the object and returns.